

FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2024
PART TEST – II
PAPER –2
TEST DATE: 10-12-2023

Time Allotted: 3 Hours

Maximum Marks: 180

General Instructions:

- The test consists of total 51 questions.
- Each subject (PCM) has 17 questions.
- This question paper contains **Three Parts**.
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Three Sections: Section-A, Section-B & Section-C**.
Section – A (01 – 04, 18 – 21, 35 – 38): This section contains **TWELVE (12)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.
Section – A (05 – 07, 22 – 24, 39 – 41): This section contains **NINE (09)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
Section – B (08 – 13, 25 – 30, 42 – 47): This section contains **EIGHTEEN (18)** numerical based questions. The answer to each question is a **NON-NEGATIVE INTEGER VALUE**.
Section – C (14 – 17, 31 – 34, 48 – 51): This section contains **SIX (06) paragraphs**. Based on each paragraph, there are **TWO (02)** questions of numerical answer type. The answer to each question is a **NUMERICAL VALUE (XXXXX.XX)**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

MARKING SCHEME

Section – A (Single Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–1	In all other cases.

Section – A (One or More than One Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen;
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen;
Partial marks	:	+2	If three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–2	In all other cases.

Section – B: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If ONLY the correct integer is entered;
Zero Marks	:	0	Question is unanswered;
Negative Marks	:	0	In all other cases.

Section – C: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct integer is entered;
Zero Marks	:	0	Question is unanswered;
Negative Marks	:	0	In all other cases.

Physics

PART – I

SECTION – A

(One Options Correct Type)

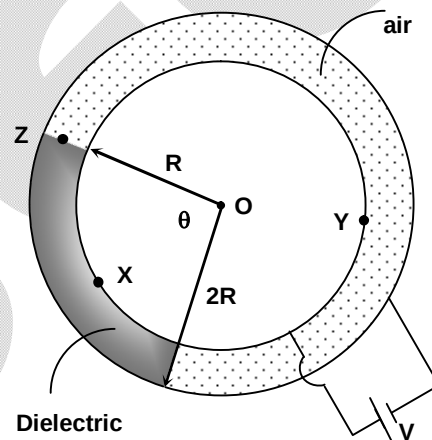
This section contains **FOUR (04)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.

- A charged ball is falling vertically under gravity along axis of fixed conducting horizontal loop. Choose the correct option. g is acceleration due to gravity.

(A) Acceleration of ball will be less than g during its motion.
 (B) Acceleration of ball be less than g when ball is above ring and equal to g when ball is below the ring.
 (C) Acceleration of ball will be equal to g when it is above ring and less than g when below the ring
 (D) Acceleration of ball will be g during its motion
- A spherical capacitor has two conducting spherical shells with radius of inner shell R and outer shell of radius $2R$. A portion of gap is filled with medium having dielectric constant K . Dielectric portion subtends angle θ at centre O . Capacitor is connected to battery of emf V . Points X and Y are on the surface of inner conducting shell. Which of following is correct?

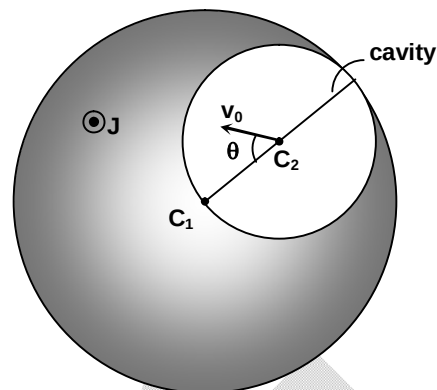
(A) surface charge density at point Y is $\left(\frac{2\epsilon_0 V}{R}\right)$
 (B) Charge density at point X is less than at point Y .
 (C) induced surface charge density at inner surface dielectric is $(1-K)\frac{2\epsilon_0 V}{R}\left(\frac{\theta}{2\pi}\right)$
 (D) Charge density at point X and Y is same
- Choose the correct option.

(A) In water lake if temperature of environment is -20°C . Equal thickness of ice layer will be formed in equal time.
 (B) If two small electric dipoles are placed in perpendicular directions at distance r , they will attract each other.
 (C) Two current carrying coil are placed co-axially having their planes parallel to each other. If one coil is tilted about its diameter, by keeping distance between their centre same, mutual inductance will remain same as $M = \sqrt{L_1 L_2}$. Where L_1 and L_2 are their self inductance.
 (D) Electric field may be conservative force field.



4. A long cylindrical wire of radius R has co-axial cylindrical cavity of radius $R/2$ touching outer surface of wire. Given wire carries current I along its axis having uniform current density over its cross section. A charge q is projected with velocity v_0 from point C_2 on axis of cavity in mid region of cavity. Velocity of charge makes angle θ with C_1C_2 and is in plane of cross section of wire when projected. Choose the correct option.

Given: $R = 2\text{ m}$, $q = 1\text{ C}$, $m = 0.2 \times 10^{-6}\text{ kg}$, $i = 3\text{ A}$, $\mu_0 = 4\pi \times 10^{-7}\text{ Tesla-meter/A}$ and $\pi^2 = 10$



- (A) For $\sin\theta = \sqrt{\frac{2}{3}}$, if velocity of charged particle is more than $\frac{1}{\sqrt{3}}\text{ m/s}$, it must collide with walls of the wire, during first half time period of its motion.
- (B) For $\sin\theta = \sqrt{\frac{2}{3}}$, v_0 should be less than $\frac{1}{4}\text{ m/s}$ to avoid collision of charged particle with walls of the wire during first time period of its motion.
- (C) For any value of v_0 charged particle will collide with walls of cylindrical wire during one time period of its motion
- (D) Information is not sufficient to determine collision of charged particle walls of cylinder.

SECTION – A

(One or More than one correct type)

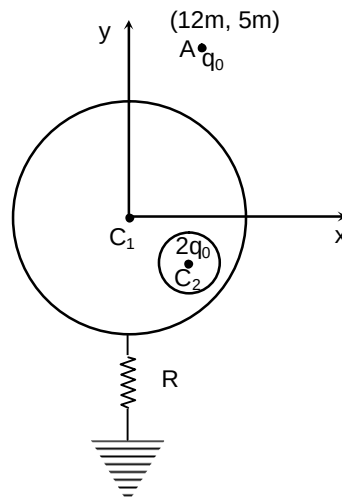
This section contains **THREE (03)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

5. A metallic sphere of mass 1 kg is heated with the help of heater. Temperature of sphere rises at constant rate 2°C/s till it reaches at 40°C . Taking atmospheric temperature 20°C . Assuming each part of sphere is at same temperature at given time and specific heat capacity of sphere is $0.5\text{ cal/gm}^\circ\text{C}$. Assuming sphere to perfect black body. Initially sphere was at room temperature. Given $1\text{ cal} = 4.2\text{ J}$, $\sigma = 5.6 \times 10^{-8}\text{ J/m}^2\text{ K}^2\text{-s}$

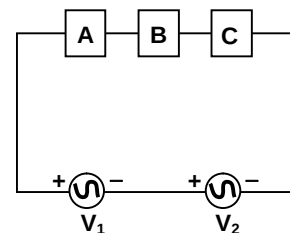
- (A) Heat absorbed by sphere till it reached at 40°C is 10 cal
- (B) Heat is being supplied to sphere at constant
- (C) Rate at which heat is being supplied to sphere is increasing with time
- (D) Rate of heat supplied to sphere can not be determined with given data

6. A very small spherical cavity is made inside a solid conducting sphere of radius 4 m having its centre C_1 at $(0, 0)$ centre C_2 of cavity is at a distance 3.25 m from C_1 . A charge $2q_0$ is placed at C_2 and another charge q_0 is placed at point A $(12\text{ m}, 5\text{ m})$. Solid sphere is connected to earth through resistance R as shown. Choose correct option(s).

- (A) Work done to move charge q_0 on circular path around C_1 with constant angular velocity, is zero.
- (B) Potential due to charge at C_2 and charge on surface of cavity is zero at position $(12\text{ m}, 5\text{ m})$
- (C) Potential at C_1 due to induced charge on outer surface of sphere is $-\frac{kq_0}{13}$ when charge q_0 is at point A
- (D) If charge q_0 is moved towards C_1 slowly potential of solid sphere will be unchanged.



7. Three boxes A, B and C in circuit contain elements R, C and L. Boxes can contain either of elements or combinations of them. Two A.C. voltages V_1 and V_2 are also connected in series as shown in circuit. Given $V_1 = 3\sin(100t + \pi/3)$ and $V_2 = 3\sqrt{3}\sin(100t - \pi/6)$. Choose the correct option(s).



- (A) If current in circuit is $i = 2 \cos(100t)$ impedance of circuit is 3Ω
 (B) If current in circuit is $i = 3 \cos(100t)$ there is no resistance in either of boxes.
 (C) if current in circuit $i = 3 \cos(100t)$, there is no capacitor and resistance in either of boxes
 (D) if current in circuit is $i = 4 \sin(100t)$ and box A contains resistance only and box B and C may contain capacitor and inductor only

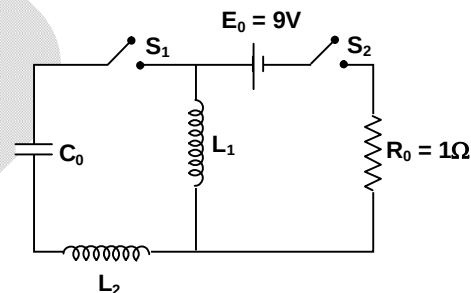
SECTION – B

(Numerical Answer Type)

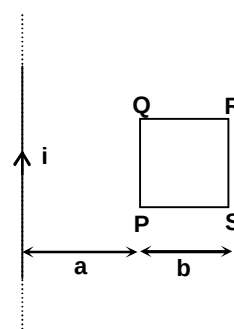
This section contains **SIX (06)** Numerical based questions. The answer to each question is a **NON-NEGATIVE INTEGER VALUE**.

8. A thin spherical shell of radius R and thickness d ($d \ll R$) is filled with liquid at temperature 40°C . Walls of container ruptures when temperature of liquid is raised to 76°C . Bulk modulus of liquid is B_L and that of shell material is B_S ($B_S = 2B_L = B_0$). The breaking stress of shell is $\frac{x\gamma B_0 R}{d}$. Find x (Ignore thermal expansion of shell material. Coefficient of volume expansion of liquid is γ .)

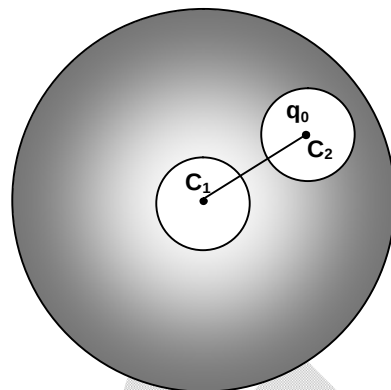
9. A capacitor of capacitance C_0 and two inductors $L_1 = L_0$ and $L_2 = 2L_0$ are connected in circuit. Initially switch S_2 is closed and S_1 is open. After long time switch S_2 is opened and S_1 is closed, both instantly. Rate of storage of energy in capacitor will be maximum when current in circuit is $\frac{x}{\sqrt{2}}$ A. Find value of x .



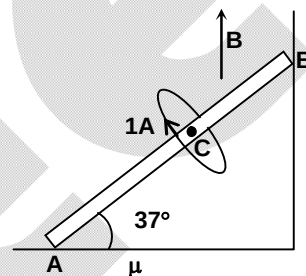
10. A square loop of resistance R and side b is placed having its side parallel to long current carrying wire and both in plane of the paper. Frame is now rotated by 180° about its side PS in time π seconds. The average torque on loop during the process is $\frac{\mu_0^2 i^2 b^2}{x\pi^2 a^2 R}$ N-m. (given $b \ll a$). Find x



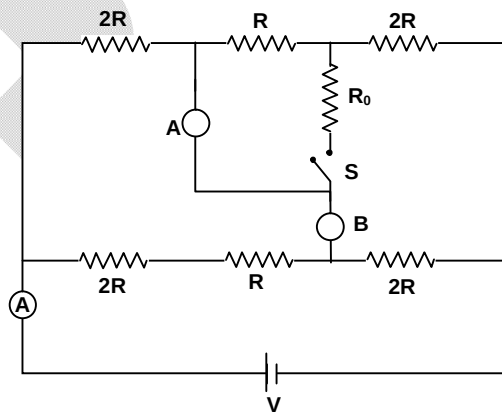
11. Two cavities with their centres at C_1 and C_2 both having radii $R/4$, are made inside solid conducting sphere of radius R as shown in the figure ($C_1C_2 = 3R/5$). A charge q_0 is placed at C_2 . The value of $|W_1 + W_2|$ is $\frac{xq_0^2}{8\pi\epsilon_0 R}$. Where, W_1 is work done in putting charge q_0 slowly from C_2 to C_1 and W_2 is work done in moving charge q_0 slowly to infinity from C_2 . Find x .



12. A ring of radius $\frac{1}{\sqrt{\pi}}$ m and of negligible mass is rigidly fixed at C by non conducting rigid spokes on non conducting rod of length 5 m and mass 2 kg. Rod being axis of ring and current of 10 A is flowing in the ring as shown in the figure. Rod resting against smooth vertical wall while its other end is on rough horizontal surface. Whole arrangement is placed in uniform vertical magnetic field of 2T. Given $AC = 4$ m, $g = 10$ m/s². Minimum value of μ so that whole system remain in equilibrium is $\frac{2x}{15}$. Find x .



13. An ideal ammeter and two nearly ideal volt meter are connected in circuit as shown with switch S is open. Resistance of voltmeter A is twice of resistance of voltmeter B. Resistance of voltmeter B is R_0 . When S is closed reading of voltmeter A changes by $\frac{xV}{75}$. Find the value of x .



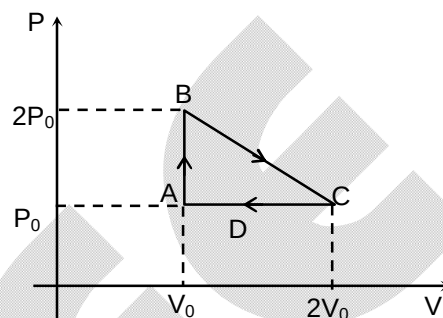
SECTION – C

(Numerical Answer Type)

This section contains **TWO (02) paragraphs**. Based on each paragraph, there are **TWO (02)** questions of numerical answer type. The answer to each question is a **NUMERICAL VALUE (XXXXX.XX)**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

Paragraph for Question Nos. 14 and 15

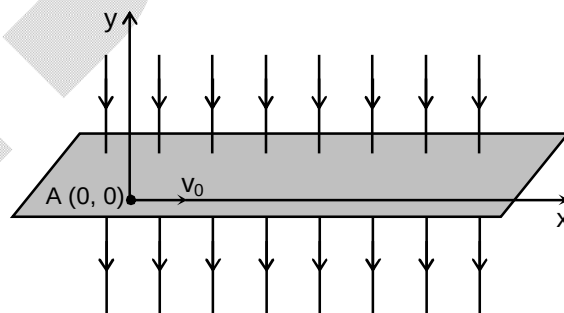
One mole of monoatomic ideal gas is taken along cyclic process ABCA. Process AB is isochoric while process AC is isobaric



14. Efficiency of cyclic process is
15. If pressure at point C is $\frac{P_0}{2}$ and volume is $2V_0$ by keeping volume at A V_0 and pressure B $2P_0$ then for the cyclic process $C \rightarrow A \rightarrow B$, heat absorbed by gas is $\times 10^5$ Joule.
(given $P_0 = 2 \times 10^5 \text{ N/m}^2$ and $V_0 = 3 \text{ m}^3$)

Paragraph for Question Nos. 16 and 17

A horizontal surface in xy-plane applies friction force on charge moving on it which is given by $f = -b_0 v$ where b_0 is positive constant and v is instantaneous velocity of charge. A charged particle is projected on rough horizontal surface with velocity v_0 at point A(0, 0) along x-axis. A vertical uniform magnetic field B_0 also exists over the surface. Let particle at $t = 0$ is at point A on surface.



16. A particle having charge q_0 and mass m_0 is given velocity v_0 at A horizontally, covers distance 2 cm in time t_0 . Another particle having charge $2q_0$, mass $2m_0$ is given velocity v_0 horizontally at A it will cover a distance ofcm in time $2t_0$.
17. A particle having $q_0 = 2 \text{ mC}$ and $m = 1 \text{ gm}$ is projected from point A on rough horizontal surface with speed $\sqrt{2} \text{ m/s}$ along x-axis. For surface $b_0 = 2 \times 10^{-3} \text{ N-s/m}$. Then distance of particle from A when its velocity become zero along x-axis ism (Given $B_0 = 1 \text{ T}$)

Chemistry

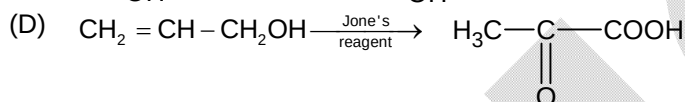
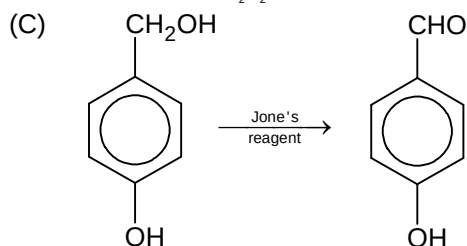
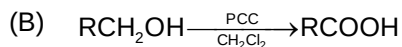
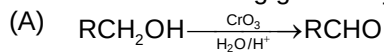
PART – II

SECTION – A

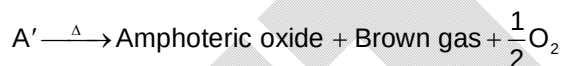
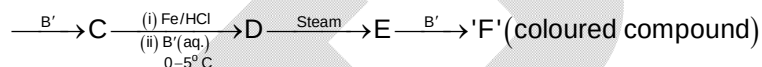
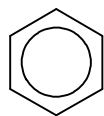
(One Options Correct Type)

This section contains **FOUR (04)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.

18. Which of the following given major product(s) are correct?



19. $\text{ZnCl}_2 (\text{anhydrous}) + \text{N}_2\text{O}_4 (\text{liq}) \longrightarrow \text{A}' + \text{B}'$

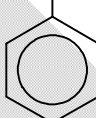


Which of the following is correct?

(A) B' is NO_2Cl

(B) 

C is



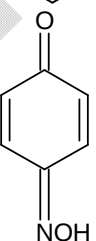
(C)

D is

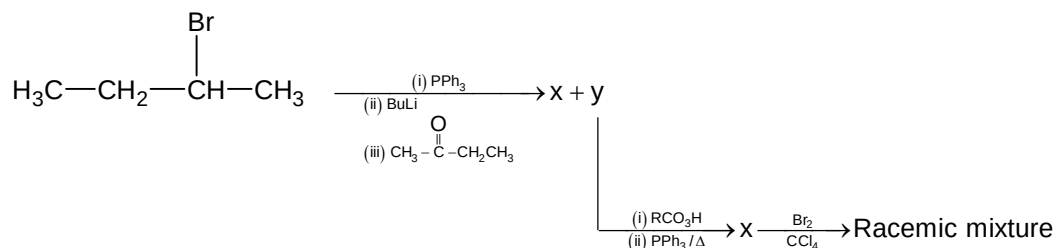


(D)

D is



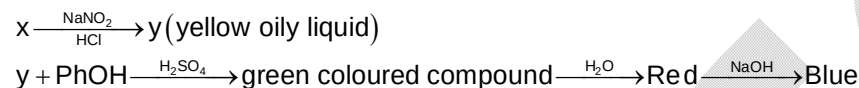
20.



Which of the following is correct?

- (A) x is trans alkene
 (B) x gives optically inactive compound after reaction with Baeyer's reagent
 (C) x gives optically active compound after hydrogenation with Ni
 (D) x is optically active compound

21.



Which of the following is correct?

- (A) x reacts with Hinsberg reagent and the product formed is dissolved in KOH
 (B) x oxidized with KMnO_4 to give tetraalkyl hydrazine
 (C) x gives carbylamine test
 (D) x can be prepared by Gabriel phthalimide reaction

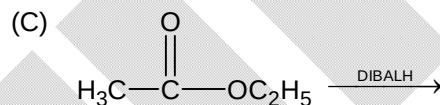
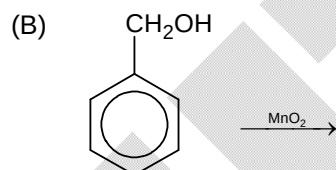
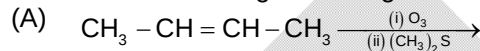
SECTION – A

(One or More than one correct type)

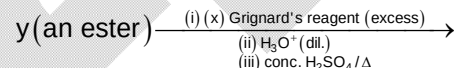
This section contains **THREE (03)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

22.

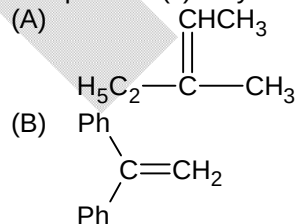
Which of the following reaction gives Aldehyde or/and Ketone as product

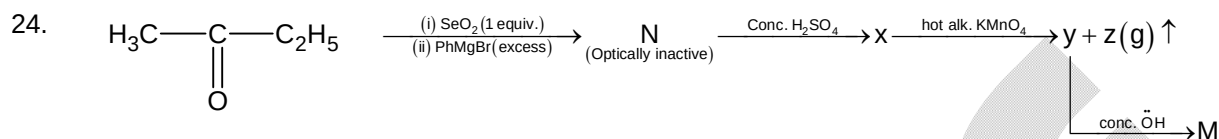
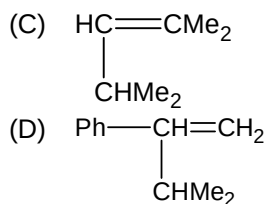


23.



The product (s) may be:





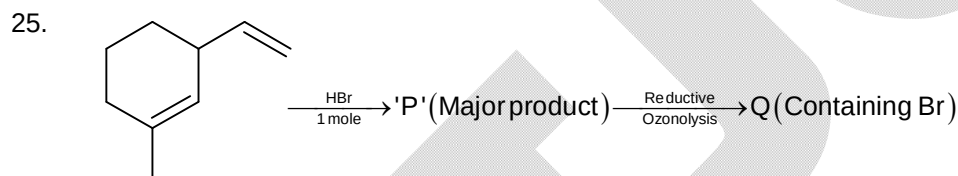
Which of the following statement (s) is/are correct?

- (A) x is a pinacol
 (B) y is an ' α ' diketone
 (C) M is optically active compound
 (D) M on heating produce an optically inactive compound.

SECTION – B

(Numerical Answer Type)

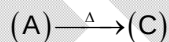
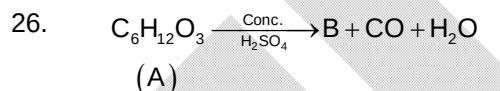
This section contains **SIX (06)** Numerical based questions. The answer to each question is a **NON-NEGATIVE INTEGER VALUE**.



'1 mole'

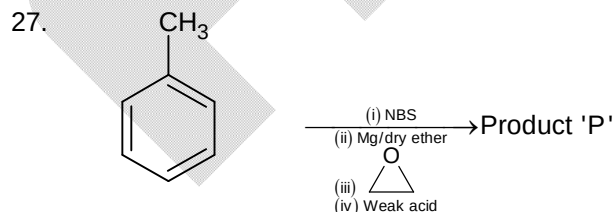
What is the molecular weight of Q?

[Given atomic wt. of C = 12, H = 1, O = 16, Br = 80]



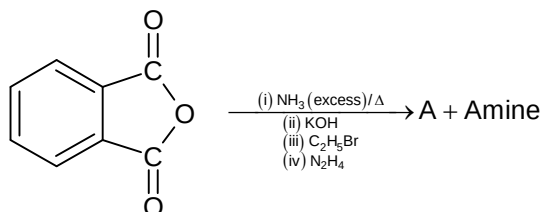
What is the molecular wt. of 'C'?

[Given atomic wt. of C = 12, O = 16, H = 1]



How many monochloroderivative will be formed by 'P' (including stereoisomers)

28.



What is the weight of atoms of 'N' present in 1 mole of A.

29. The pK_a of ' $-\text{COOH}$ ' group in an amino acid is 3.59 and the pK_a of conjugate acid of amino group of the same amino acid is 8.41. Then what will be the isoelectric point of the same amino acid?
30. α -Anomer of a carbohydrate has specific angle of rotation 111° and β -Anomer of same carbohydrate has specific angle of rotation equal to 20 at equilibrium between α & β -anomer the net angle of rotation is 78.24. Then calculate the % of α -Anomer at equilibrium.

SECTION – C (Numerical Answer Type)

This section contains **TWO (02) paragraphs**. Based on each paragraph, there are **TWO (02)** questions of numerical answer type. The answer to each question is a **NUMERICAL VALUE (XXXXX.XX)**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

Paragraph for Question Nos. 31 and 32

Reactivity ratio of $3^\circ\text{H} : 2^\circ\text{H} : 1^\circ\text{H} = 5 : 3.8 : 1$ for chlorination of alkanes.

31. Calculate percentage of the least probability factor product, formed by the chlorination of isohexane
32. Product obtained by the monochlorination of isohexane (having maximum probability factor) is _____ %

Paragraph for Question Nos. 33 and 34



Y contain asymmetric 'C' atoms

Z does not show geometric isomerism

P gives positive Schiff's test but N does not

33. What is the molecular weight of Y?
[Given atomic wt. of O = 16, C = 12, Br = 79.9]
34. $\text{N} + \text{Cl}_2 / \text{OH}^- \longrightarrow \text{Q} + \text{Salt of cabroxylic acid}$
What is the molecular weight of Q?

Mathematics**PART – III****SECTION – A****(One Options Correct Type)**

This section contains **FOUR (04)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.

35. A tangent is drawn at any point P on the parabola $y^2 = 32x$. Now two mutually perpendicular tangents are drawn to the circle $x^2 + y^2 = 32$ from any point on this tangent. If the chord of contact of the circle and tangent at P always meet in fixed point, then possible coordinates of P is/are
 (A) (8, 16), (8, -16) (B) (2, 8) and (2, -8)
 (C) $\left(\frac{1}{2}, 4\right), \left(\frac{1}{2}, -4\right)$ (D) $\left(\frac{1}{8}, 2\right), \left(\frac{1}{8}, -2\right)$
36. Let $x = \cot^{-1}\left(\frac{31}{25}\right) + \cot^{-1}\left(\frac{91}{25}\right) + \cot^{-1}\left(\frac{201}{25}\right) + \cot^{-1}\left(\frac{361}{25}\right) + \dots \infty$, then $15 \cot x$ is
 (A) 2 (B) 3
 (C) 4 (D) 5
37. Consider an ellipse $E: \frac{x^2}{25} + \frac{y^2}{80} = 1$ and a chord L of the ellipse E which is bisected at point $P\left(\frac{5}{2}, \alpha\right)$; $\alpha \in I$. If the chord has integral slope and P_1 and P_2 be the position of P for which $|\alpha|$ is maximum, then
 (A) number of possible values of α is/are 7
 (B) number of possible values of α is/are 8
 (C) equation of parabola having latus rectum P_1P_2 is $y^2 = -8\left(x - \frac{9}{2}\right)$
 (D) the chord passing through the points P_1 and P_2 intersect at $\left(\frac{11}{2}, 0\right)$
38. In an isosceles triangle ABC with $AB = AC$, the bisector of angle C meets side AB at D, if $BC = AC + AD$, then
 (A) $\angle A = \frac{\pi}{3}$
 (B) $\angle C = \frac{3\pi}{8}$
 (C) length of angle bisector $CD = (4 - 2\sqrt{2})AB \cos \frac{\pi}{8}$
 (D) length of angle bisector $CD = (4 + 2\sqrt{2})AB \cos \frac{\pi}{8}$

SECTION – A

(One or More than one correct type)

This section contains **THREE (03)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

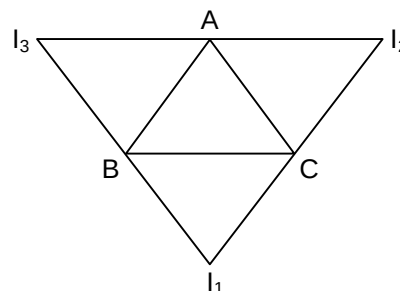
39. The normal at $P_1(4, 2)$ on the curve $xy = 8$ meets the curve again at P_2 , then normal at P_2 meets the curve again in P_3 and so on, then
 (A) slope of normal at P_{100} is $2^{3^{99}}$
 (B) slope of normal at P_{100} is $\frac{1}{2^{3^{99}}}$
 (C) if chord joining P_1 and P_4 subtends right angle at point Q on the curve, then slope of normal at Q is 2^{13}
 (D) slope of tangent at P_9 is $\frac{1}{2^{3^8}}$
40. A hyperbola having eccentricity $\frac{5}{4}$ passes through the point $P(5, 6)$. If the nearer focus is $(2, 2)$ and equation of tangent at P on the hyperbola is $3x + 4y - 39 = 0$, then
 (A) length of latus rectum of hyperbola is 22.5
 (B) length of chord through $(2, 2)$ parallel to the tangent at P is 22.5
 (C) length of perpendicular from the focus other than $(2, 2)$ on the tangent at P is 45
 (D) area of the triangle enclosed by the tangent at P and asymptotes of the hyperbola is 300
41. Let $f(x) = \cos^{-1} \cos\left(\frac{24 + 4x^2}{4 + x^2}\right)$. If $f(x) + 5 - 2\pi \leq 0$, then
 (A) number of integral points (x, y) lying on the line $9x + \alpha^2 y = 16$; $\alpha \in I$ is/are 8
 (B) area bounded by $y = f(x)$ and x-axis is $6\pi - 16$
 (C) range of $f(x)$ is $[2\pi - 6, 2\pi - 4)$
 (D) range of $f(x)$ is $[2\pi - 5, 2\pi - 4)$

SECTION – B

(Numerical Answer Type)

This section contains **SIX (06)** Numerical based questions. The answer to each question is a **NON-NEGATIVE INTEGER VALUE**.

42. Consider a parabola $y^2 = 4x$ and an ellipse $\frac{x^2}{50} + \frac{y^2}{4} = 1$. Number of points $P(h, k)$; $h, k \in I$ inside the ellipse from which exactly one normal can be drawn to the parabola is/are
43. Consider triangle ABC with $\angle B = \angle C = \frac{2\pi}{5}$. Let $l_1 l_2 l_3$ be the triangle formed by joining the external bisectors of $\angle A$, $\angle B$ and $\angle C$ as shown in the figure and $\triangle DEF$ be the triangle formed by joining the feet of perpendicular from the vertices of $\triangle ABC$ on opposite sides. If $l_1 l_2 = 4(\sqrt{5} + 1)$ and area of $\triangle DEF$ is k , then the value of $[k^2]$ is (where $[.]$ represents greatest integer function)



44. The number of solution(s) of the equation $(10 + 6\sqrt{3})\sin^3 x + (6\sqrt{3} - 10)\cos^3 x + 6\sin 2x = 8$ which satisfy $|\tan x| > 1$; $x \in [0, 2023\pi]$ is/are
45. Let a point P lies on $5x - 12y - 17 = 0$ and point Q lies on $12x + 5y - 7 = 0$ and point R is $(-4, 11)$. If PQRS is a rhombus having side length 13 units, then locus of S is a conic whose latus rectum is k units, then the value of k^2 is
46. If $\tan\left(x + \frac{\pi}{9}\right) = \tan\left(x + \frac{\pi}{18}\right)\tan x \tan\left(x - \frac{\pi}{18}\right)$ has two distinct solutions α and β in $\left(0, \frac{\pi}{2}\right)$; $\alpha < \beta$, then $4\tan^2\left(\frac{7\alpha}{6} + \beta\right)$ is equal to
47. Consider a circle $S = 0$ which touches the line $3x + 4y - 5 = 0$ at $P(-5, 5)$ and intercepts a chord of length $2\sqrt{161}$ on the line $4x - 3y - 5 = 0$. Also $(-1, -1)$ lies inside the circle. Let the point on the circle $S = 0$ at a distance of $\sqrt{90}$ units from the point $P(-5, 5)$ be (α, β) ; $\alpha > 0$, then the value of $(\alpha - 2\beta)$ is

SECTION – C

(Numerical Answer Type)

This section contains **TWO (02) paragraphs**. Based on each paragraph, there are **TWO (02)** questions of numerical answer type. The answer to each question is a **NUMERICAL VALUE (XXXXX.XX)**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

Paragraph for Question Nos. 48 and 49

A circle having centre C lying on the curve $\frac{x^2}{9} - \frac{y^2}{4} = 1$ intersects the rectangular hyperbola $xy = 72$ in four points (x_i, y_i) ; $i = 1, 2, 3, 4$ represented by P, Q, R, S

48. If C is fixed, then locus of centroid of triangle PQR is a conic whose latus rectum is of length
49. If C is variable, then locus of focus of the locus of centroid of ΔPQR is a conic having eccentricity e where $18e^2$ is

Paragraph for Question Nos. 50 and 51

Let $a = \cos^{-1}\cos 60$, $b = \sin^{-1}\sin 50$, $c = \tan^{-1}\tan(a + b)$. If the greatest integer x for which $\cos^{-1}\cos x \geq |x + c - a - b|$ is equal to k, then

50. The value of k is
51. If $\cos^{-1}\left(\frac{3k}{10} - 4\left(\frac{k}{10}\right)^3\right) = \alpha \cos^{-1}\frac{k}{10} + \beta$, then $\alpha - \frac{2\beta}{\pi}$ is